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RESEARCH ARTICLE



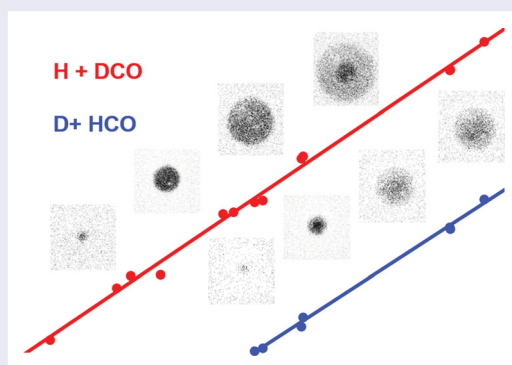
HDCO radical dissociation thresholds by velocity map imaging

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ABSTRACT

Ground state radical dissociation pathways of HDCO were investigated. Frequencies of rovibrational lines from a variety of vibrational bands of the first singlet excited state over a broad excitation energy range were determined and subsequently used to excite singly deuterated formaldehyde, HDCO. Translational energy release of H and D atoms measured by velocity map imaging was then used to determine accurate threshold energies of $30,448 \pm 5 \text{ cm}^{-1}$ for $\text{H} + \text{DCO}$ and $30,949 \pm 3 \text{ cm}^{-1}$ for $\text{D} + \text{HCO}$.



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Introduction

Formaldehyde has been vital to advancements in spectroscopy and the understanding of polyatomic dissociation dynamics [1–16]. The simplest four atom molecule, it is tractable theoretically and easily studied via absorption of frequency doubled light from YAG-pumped dye lasers for electronic spectroscopy [2,17–20]. Infrared and microwave radiation have also been used for thorough vibrational and rotational investigations, respectively [21–29]. Accessible and long-lived singlet and triplet excited states, and radical and molecular dissociation pathways in formaldehyde make it a perfect prototype system to develop an understanding of various dynamical events. A characterisation of the roaming mechanism was first provided through joint theoretical and experimental studies of H_2CO [13,14,30]. Through these studies, another pathway to molecular products, in addition to the known conventional, tight, three-centre transition state was detailed that involved extensive H atom

roaming leading to intramolecular reaction between H and HCO [14].

Investigation of the isotopologues D_2CO and HDCO and comparison to H_2CO has improved spectroscopic theory and provided general insight into molecular dissociation dynamics [31–35]. Formaldehyde is also one of the most abundant molecules in interstellar space and HDCO has been detected in OMC-1, cometary objects, and protostars [36,37]. Observation of deuterated species may help determine the physical and chemical conditions of star-forming regions [36,38–40]. Extensive rotational analyses have been performed by microwave and millimetre wave methods, including a study using several spectrometers and older measurements compared to *ab initio* predictions [23]. Infrared and Raman spectra of HDCO were first reported by Davidson, Stoicheff, and Bernstein [26]. The ground vibrational state was further characterised through analysis of rotational fine structure by Job, Sethuraman, and Innes [33], ν_3 , ν_5 and ν_6 were

investigated by Johns and McKellar using conventional infrared absorption and CO laser Stark spectroscopy [24,25], and finally, all fundamentals were updated and summarised by Clouthier and co-workers [8]. Large scale variational calculations were also performed to determine the vibrational structure and mixing [21,22], and higher resolution measurements have improved agreement between theoretical and experimental line intensities and positions in addition to new line lists over a broader frequency range [41].

In addition to ground state characterisation, Job and co-workers provided a thorough vibrational and rotational analysis of the $^1A_2-^1A_1$ system [33], including determination of excited state fundamentals. Bands of this system were later reinvestigated with higher resolution and magnetic rotation [42], leading to a reassignment of v_3 to v_6 and deduction of a mechanism for selection rules. A similar vibrational and rotational analysis of the $^3A_2-^1A_1$ system for four vibronic bands, as was done for $^1A_2-^1A_1$, determined the triplet geometry and identified Coriolis perturbations [43].

Deuterium substitution results in longer fluorescence decay times and causes the CO rotational distribution from the tight transition state pathway to be shifted higher [34,35,44,45]. Shorter lifetimes in H_2CO are ascribed to S_1 coupling to the S_0 lumpy continuum where ground state vibrational levels are broadened by tunnelling through the barrier to molecular products [2,12,34,46]. A higher CO rotational distribution can be explained using the rotational reflection principle, where the maximum j_{CO} is scaled with the square root of the reduced mass of deuterium substitution of H_2 [31,47]. A comprehensive study of H_2CO , HDCO, and D_2CO using photoacoustic absorption and VUV laser-induced fluorescence detection of atomic products then determined T_1 barrier height, the S_1-T_1 intersystem crossing rate, and radical dissociation thresholds [48]. Threshold energies were calculated assuming a Born–Oppenheimer potential, and threshold differences arise from zero-point energy differences. For H_2CO and D_2CO , lower bounds were determined experimentally by comparing photoacoustic and photofragment excitation (PHOFEX) spectra and upper bounds were set at the origin of the vibrational band where the PHOFEX and photoacoustic spectra match completely.

Decades of experimental and theoretical work have since been able to determine the radical threshold for H_2CO , $30327.6 \pm 0.9 \text{ cm}^{-1}$, accurately [31,49,50]. The same cannot be said for HDCO. For HDCO, only the D atom radical threshold has been obtained, derived from an extrapolation of the H/D branching ratio to infinity [48]. The experimental and calculated thresholds reported have a $\sim 100 \text{ cm}^{-1}$ uncertainty and have not

been updated for more than three decades. Using H and D atom REMPI (resonance-enhanced multiphoton ionisation) detection and velocity map imaging [51], translational energy of atomic fragments at a range of energies allows for a more accurate determination of HDCO ground state radical thresholds with little uncertainty.

Materials and methods

Synthesis of paraformaldehyde-d1 was adapted from a previously described method, which is detailed below [52]. $CDBr_3$ (50 g) was added to an oven-dried 200 ml round bottom flask, attached to a reflux condenser and an argon-filled balloon, and then Bu_3SnH (57 g) was added dropwise carefully. After addition and waiting for 10 min, the reaction mixture was heated to reflux for 25 h. The mixture then was distilled to give pure $CDHBr_2$. $CDHBr_2$ (15 g) was placed in a 200 mL round bottom flask. Glacial acetic acid (54 ml) and acetic anhydride (6 ml) were added. Potassium acetate (26 g) was then added in one portion and the reflux condenser and an argon-filled balloon were quickly attached to the round bottom flask. The reaction mixture was heated at reflux for 40 h. The mixture was then allowed to cool to room temperature and 150 ml of diethyl ether were added. White precipitated potassium bromide was filtered off several times and more diethyl ether was added until there was no precipitated potassium bromide formed. The filtrates were combined and the solvent was removed by a rotary evaporator to afford the pure D-methylene diacetate in 78% yield. D-Methylene diacetate (9 g) was placed in a 25 mL round bottom flask and HPLC grade water (1 ml) and conc. HCl (1 ml) were added. The reaction mixture was then heated to reflux for 24 h. The solvent was removed by oil pump and paraformaldehyde-d1 was formed as a white powder.

A brief description of the previously described [53] experimental method and apparatus used in these experiments is included here. Source and detection chambers are pumped to 3×10^{-4} and 2×10^{-7} Torr, respectively, by turbomolecular pumps and separated by an aluminium plate on which a skimmer is mounted. A pulsed solenoid valve (Parker-General) introduced the molecular beam of HDCO seeded in helium into the source chamber and was subsequently supersonically expanded and skimmed. HDCO monomers were produced by a method adapted from previous experiments in which monomers are liberated from paraformaldehyde-d1 [54]. Paraformaldehyde-d1, anhydrous $MgSO_4$, and glass wool were packed into a KF elbow before the valve. The KF elbow, valve, and connecting stainless steel parts were wrapped with braided fibreglass heating cord and heated to $\sim 70^\circ\text{C}$ to liberate HDCO monomers from

paraformaldehyde-d1. A 930 Torr helium backing pressure resulted in a $\sim 5\%$ formaldehyde concentration and a beam temperature of ~ 18 K. Beam temperature was determined from relative populations of HDCO rovibrational lines and known experimental conditions of H_2CO experiments.

UV light was generated by frequency doubled or tripled narrow linewidth (0.1 cm^{-1}) dye lasers pumped by Nd:YAG lasers. Pump energies were less than ~ 0.2 mJ/pulse and probe energies were ~ 0.2 mJ/pulse. PHOFEX spectra were obtained by scanning the excitation/dissociation laser from 30,100 to 31,800 cm^{-1} for rovibrational bands on S_1 and using REMPI detection of $j_{\text{CO}} = 44$ via $Q(\text{B}^1\Sigma^+ \leftarrow \text{X}^1\Sigma^+)$. Spectral simulations were done using PGOPHER [55] and a fit was performed to assign rotational structure in vibrational bands of the experimental spectra. HDCO was then photodissociated with different pump frequencies corresponding to different rovibrational lines. HDCO ground and excited state J were between 0 and 4, with a rotational energy typically around 6 cm^{-1} and up to 26 cm^{-1} . H and D atoms were detected by 2 + 1 REMPI via $2s(^2S_{1/2}) \leftarrow 1s(^2S_{1/2})$ near 243 nm.

Ions produced by REMPI were accelerated through a time-of-flight tube by an ion optics assembly toward a 75 mm microchannel plate detector coupled to a P-47 phosphor screen (BOS-75-OPT01, Beam Imaging Solutions) for DC slice velocity map ion imaging with a 75 ns MCP pulse width [14,51,53,56]. Images were captured using a USB CCD camera (iDS uEye UI-2230SE-M-GL) and in-house data acquisition software NuACQ. A photomultiplier tube monitoring the phosphor screen obtained time-of-flight information. The experiment was performed at 10 Hz. Image reconstruction and data extraction were done using our finite slice analysis method, FinA [57,58].

Results

Excitation energies were determined by first obtaining PHOFEX spectra of HDCO over a wide range. CO REMPI was used as the probe because molecular dissociation occurs well below both radical thresholds via a tight transition state pathway and this way background from H or D atoms from formyl multiphoton dissociation can also be avoided. A dense PHOFEX spectrum with more than twenty vibrational bands was recorded, Figure 1. The spectral density is further demonstrated over a 250 cm^{-1} range, Figure 1 inset, with at least three partially overlapped vibrational bands, 2^26^1 , 2^24^3 , and $2^24^26^1$. Green dots above the transitions indicate the rotational lines used to prepare HDCO on S_1 in the imaging experiments. The radical thresholds determined in this work are also marked in Figure 1.

Upon imaging H and D atom photoproducts using four different pump energies (Figure 2 left and right columns, respectively), features with radii that increase with energy are observed. The top image is recorded on the rovibrational line where H and D atoms were first detected, indicated by the small centre spot. The three subsequent images have pump energies increased by hundreds of wavenumbers. The increase in the maximum radius at which ions are observed with pump energy reflects the excess the pump energy above the radical threshold. Plotting the translational energy vs the upper state energy of the transition, Figure 3, the intercept allows for the radical threshold energy to be determined more accurately than simply using the lowest energy rotational line that yielded atomic fragments. H and D atom thresholds of $30,448 \pm 5$ and $30,949 \pm 3\text{ cm}^{-1}$ were thus determined. Using the latter value with the heats of formation of D and HCO from the Active Thermochemical Tables [50,59], we obtain $\Delta H_f^0(0\text{ K}) = -109.03 \pm 0.10$

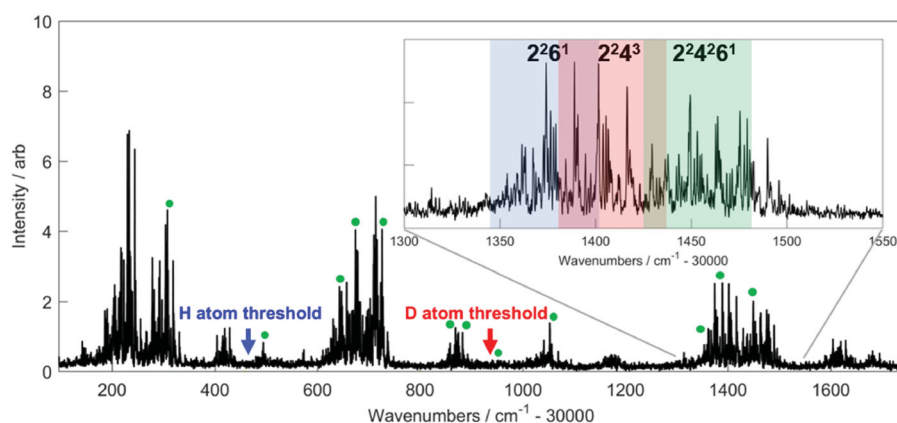


Figure 1. HDCO PHOFEX spectra with $j_{\text{CO}} = 44$ REMPI detection. Green dots indicate transitions excited in imaging experiments. Inset: expanded view of select vibrational bands.

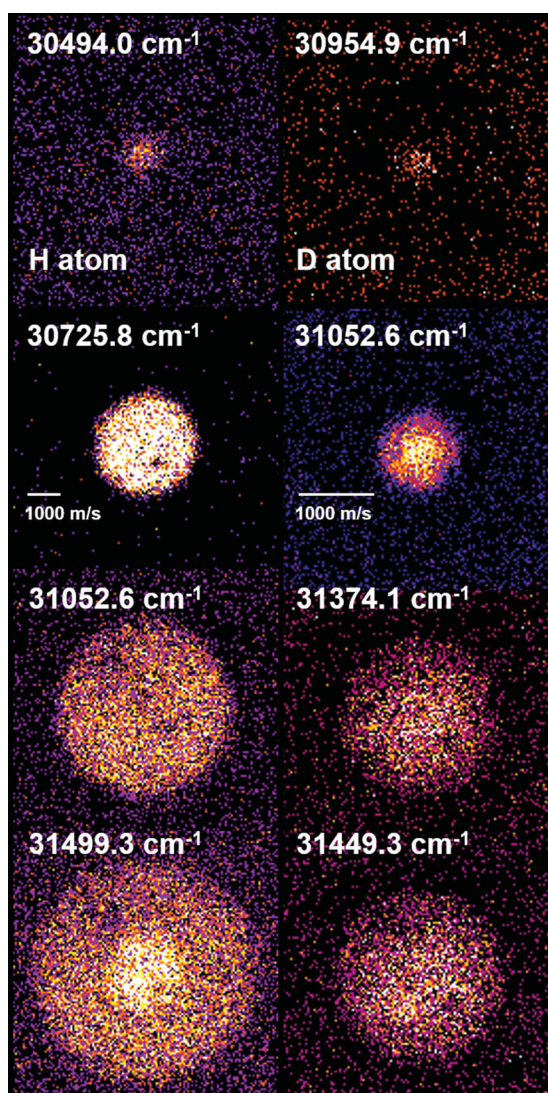


Figure 2. Selected images of H atom (left column) and D atom (right column) fragments from excitation on four different rovibrational lines of HDCO.

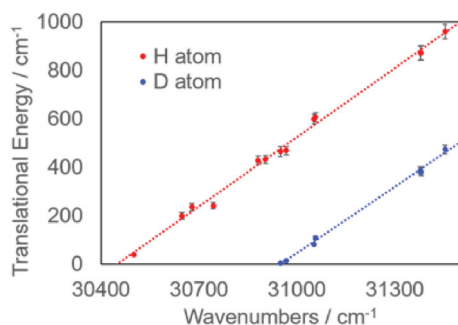


Figure 3. Translational energy plotted against upper state energy for H atom (red) and D atom (blue) fragments.

kJ/mol for HDCO. Using this with the H atom threshold we obtain $\Delta H_f^0(0\text{ K}) = 39.17 \pm 0.14$ kJ/mol for DCO.

D atom translational energy distributions, Figure 4, for three of the images in Figure 2 show the expected

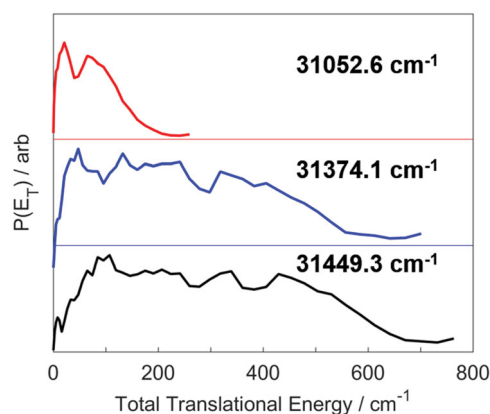


Figure 4. D atom translational energy distributions of three selected images displayed in Figure 2.

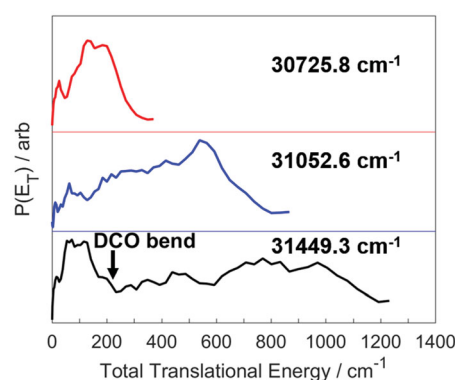


Figure 5. H atom translational energy distributions of three selected images displayed in Figure 2.

behaviour, with increased pump energy directly correlating with increased translational energy. H atom translational energy distributions, Figure 5, for three of the images in Figure 2, are not broad and structureless like the D atom $P(E)$. The H atom image from the 31,449.3 cm^{-1} excitation also exhibits a less intense faster distribution of fragments around an intense centre spot, indicating concomitant excitation of the DCO bend.

Discussion

Many S_1 vibrational bands have previously been observed, and 1A_2 – 1A_1 rovibrational lines assigned. Only select frequency ranges, however, have been assigned, likely due to spectral complexity and the relatively low resolution of previous measurements. Here, we have recorded PHOFEX spectra, Figure 1, over an entire region at energies spanning nearly 2000 cm^{-1} . The density of rovibrational lines and overlapping vibrational bands is exhibited by different coloured boxes highlighting three different vibrational bands in Figure 1 inset. A detailed analysis and assignment of these vibrational bands will be reported later.

Previously published calculated and experimental thresholds (often calculated for the different isotopologues using vibrational frequencies to predict the difference) were at least 35 cm^{-1} lower than H and D atom thresholds of $30,448 \pm 5$ and $30,949 \pm 3\text{ cm}^{-1}$ determined here. A linear least squares analysis was performed to determine the intercepts and the standard error of the dependent variable was used to determine the uncertainty. Uncertainty in these thresholds arises principally from the determination of the radius. Radii of images were determined by averaging the radius obtained in each quadrant separately, resulting in a standard deviation between 4 and 30 cm^{-1} for each point. Larger standard deviations at higher energies reflected in the error bars of Figure 3 that contain two standard deviations, are a consequence of uniform uncertainty in image radii giving a quadratic dependence of the uncertainty in energy. This study offers the first reported experimental value for the H atom ground state radical threshold. As for the D atom threshold, we note that at a pump energy of $30,885.62\text{ cm}^{-1}$, an energy near the upper bound of the measured and calculated threshold range previously reported, D atoms were no longer detected. The previous experimental determination of the D atom threshold derives from a rough extrapolation of the H/D branching ratio [48]. While the new value lies above the upper limit of the range determined in that study, the measurement here of translational energy release from a range of excitation energies has the advantage of allowing for a direct determination.

Excitation of HDCO at all energies resulted in relatively uniform D atom translational energy distributions, Figure 4. Those for H atom peaked at a higher energy corresponding to its nearly 500 cm^{-1} lower threshold, as is reflected in the $P(E)$ plots from $31,052.6\text{ cm}^{-1}$ excitation in Figures 4 and 5. The lower H atom threshold also results in more available energy for its cofragment DCO. At the highest excitation energy used, $31,449.3\text{ cm}^{-1}$, the H atom $P(E)$ in Figure 5 decreases with increasing translational energy beyond 140 cm^{-1} then rises again above $\sim 200\text{ cm}^{-1}$. A total translational energy of 140 cm^{-1} corresponds to a DCO internal energy of 866 cm^{-1} , closely matching the frequency of the $\sim 850\text{ cm}^{-1}$ DCO bending mode [60–65]. All other vibrational modes are higher in energy and just beyond the available energy. The HCO bending mode, $\sim 1080\text{ cm}^{-1}$, is not seen in the D atom $P(E)$ in Figure 4 because the excitation energy is insufficient. The H atom $P(E)$, particularly at high energy beyond the DCO bending mode, does not appear to be purely statistical. Future studies are necessary to investigate the underlying dynamics and potential involvement of the T_1 barrier. Previously, the distribution of population in HCO states characterised by different values of K_c

was sensitive to initially prepared K_c of H_2CO [54]. Oscillations in detailed HCO product state distributions over a more than 300 cm^{-1} range could not be explained, showing deviations from calculations based on phase space theory when K_a and K_c were included [66].

PHOFEX spectra over a wide range will allow for the assignment of more vibrational bands and determination of spectroscopic constants. Known frequencies of rotational lines can be used to study the dynamics of different vibrational bands and the energy dependence of different processes. Accurate ground state radical thresholds obtained from velocity map imaging will assist future HDCO dynamics studies. Threshold phenomena such as competing roaming and non-roaming molecular pathways that occur in H_2CO require exact threshold energies and an understanding of the underlying dynamics in the radical dissociation channel. The effect on HDCO of excitation at each atomic threshold and the energies in between each may reveal essential information on the roaming mechanism and assist in refining the PESs.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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